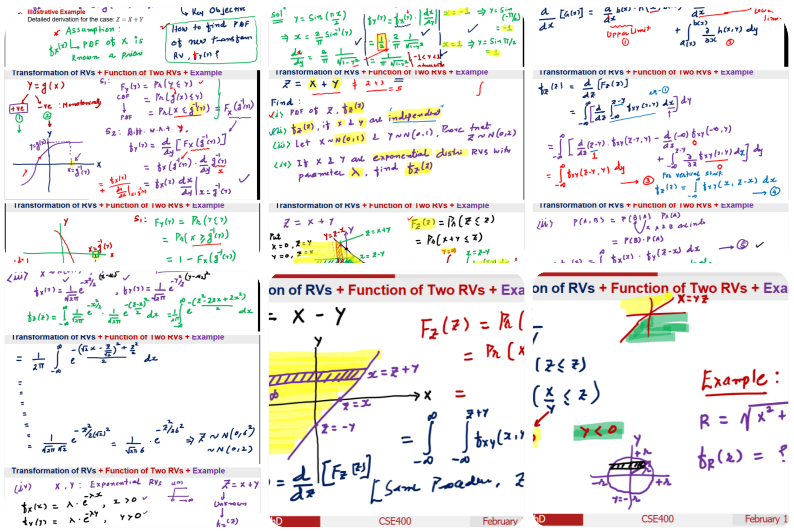
Transformation of RVs



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Lecture Scribe

Transformation of Random Variables + Function of Two Random Variables + Example

CSE400 – February 10, 2026

Dhaval Patel, PhD

---

1. Transformation of Random Variables

1.1 Learning Objective

The lecture begins with:

> **Transformation of Random Variables**
> Learning of transformation techniques for random variables.

The key objective highlighted in the slides:

> **How to find PDF of new transformed RV, $f_Y(y)$?**

Assumption explicitly stated:

$$f_X(x) \rightarrow \text{PDF of } X \text{ is known a priori}$$

---

2. Function of One Random Variable

2.1 Setup

Let:

$$Y = g(X)$$

The slides distinguish two cases:

- (+ve) Monotonically increasing
- (–ve) Monotonically decreasing

The derivation is performed separately for each case.

2.2 Case 1: $g(x)$ Monotonically Increasing

Step S1: Compute CDF of Y

$$F_Y(y) = \Pr(Y \leq y)$$

Since $Y = g(X)$:

$$= \Pr(g(X) \leq y)$$

Because g is increasing:

$$= \Pr(X \leq g^{-1}(y))$$

Thus:

$$F_Y(y) = F_X(g^{-1}(y))$$

Step S2: Differentiate w.r.t. y

$$f_Y(y) = \frac{d}{dy}\left[F_X(g^{-1}(y))\right]$$

Using differentiation:

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Equivalently written as:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

2.3 Case 2: $g(x)$ Monotonically Decreasing

Step S1: CDF

$$F_Y(y) = \Pr(Y \leq y)$$
$$= \Pr(X \geq g^{-1}(y))$$
$$= 1 - F_X(g^{-1}(y))$$

Step S2: Differentiate

The slide shows:

$$f_Y(y) = f_X(x) \left| \frac{dy}{dx} \right|^{-1}_{x=g^{-1}(y)}$$

The final boxed formula (valid for both +ve and –ve cases):

$$\boxed{f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|_{x=g^{-1}(y)}}}$$

Step S3 (explicit note in slide):

> Change the limits for y

---

3. Example: Single Variable Transformation

3.1 Given

$$X \sim \text{Uniform}(-1, 1)$$

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

PDF of X:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

3.2 Solution

From:

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Invert:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Differentiate:

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

Using transformation formula:

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

$$= \frac{1}{2}\cdot\frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}$$

With limits:

$$-1 < y < 1$$

---

4. Function of Two Random Variables

4.1 Setup

$$Z = X + Y$$

Tasks shown:

1. Find PDF of Z, $f_Z(z)$
2. If X and Y are independent
3. If $X \sim N(0,1)$, $Y \sim N(0,1)$, prove $Z \sim N(0,2)$
4. If X, Y exponential with parameter λ, find $f_Z(z)$

---

5. Derivation of PDF of $Z = X + Y$

5.1 CDF Approach

$$F_Z(z) = \Pr(Z \leq z)$$
$$= \Pr(X + Y \leq z)$$

Region described by:

$$x \leq z - y$$

Thus:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

5.2 Leibniz Rule (as shown in slide)

If:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

Then:

$$\frac{d}{dx} G(x) = \frac{d}{dx} b(x)\, h(x, b(x)) - \frac{d}{dx} a(x)\, h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)\, dy$$

5.3 Apply to Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$= \int_{-\infty}^{\infty} f_{XY}(z-y, y)\, dy$$

Equivalently (vertical strip):

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z-x)\, dx}$$

---

6. Independent Case

If X and Y independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx}$$

Slide labels this:

> **Convolution Integral**

---

7. Normal Example

Given:

$$X \sim N(0,1), \quad Y \sim N(0,1)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

After algebra shown in slides:

$$f_Z(z) = \frac{1}{\sqrt{2\pi \cdot 2}} e^{-z^2/(2\cdot 2)}$$

Thus:

$$\boxed{Z \sim N(0,2)}$$

---

8. Exponential Example

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} u(x) u(z-x)\, dx$$

Where:

$$u(x)u(z-x) = \begin{cases} 1, & 0 < x < z \\ 0, & \text{otherwise} \end{cases}$$

Thus integration reduces to:

$$\begin{aligned} f_Z(z) &= \lambda^2 e^{-\lambda z} \int_0^z dx \\ &= \lambda^2 z e^{-\lambda z} \end{aligned}$$

---

9. Example: $Z = X - Y$

$$\begin{aligned} F_Z(z) &= \Pr(Z \leq z) \\ &= \Pr(X - Y \leq z) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y)\, dx\, dy \end{aligned}$$

Then:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

(Same procedure as $Z = X + Y$)

---

10. Ratio Example

$$Z = \frac{X}{Y}$$

$$F_Z(z) = \Pr\left(\frac{X}{Y} \leq z\right)$$

Cases shown:

- $Y > 0$
- $Y < 0$

Region drawn accordingly.

---

11. Radial Example

$$R = \sqrt{X^2 + Y^2}$$

Find:

$$f_R(r) = ?$$

(Example posed on slide.)

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LECTURE SCRIBE

Transformation of Random Variables + Function of Two Random Variables + Example

CSE400 – February 10, 2026

Instructor: Dhaval Patel, PhD

---

1. Transformation of Random Variables

1.1 Learning Objective

The lecture begins with the topic:

> **Transformation of Random Variables**
> Learning of transformation techniques for random variables.

The slide explicitly highlights the **Key Objective**:

$$\textbf{How to find PDF of new transformed RV, } f_Y(y)\textbf{?}$$

An assumption is written on the slide:

$$\textbf{Assumption: } f_X(x) \rightarrow \text{PDF of } X \text{ is known a priori}$$

Thus, throughout the derivation, the PDF of X is assumed known.

---

2. Function of One Random Variable

2.1 Setup

Let:

$$Y = g(X)$$

The slide distinguishes two cases:

- (+ve) Monotonically increasing
- (–ve) Monotonically decreasing

The derivations are performed separately for both cases.

2.2 Case 1: $g(x)$ Monotonically Increasing

Step S1: Compute CDF of Y

$$F_Y(y) = \Pr(Y \leq y)$$

Since $Y = g(X)$:

$$= \Pr(g(X) \leq y)$$

Because $g(x)$ is monotonically increasing:

$$= \Pr(X \leq g^{-1}(y))$$

Therefore:

$$F_Y(y) = F_X(g^{-1}(y))$$

Step S2: Differentiate w.r.t. y

$$f_Y(y) = \frac{d}{dy}\left[F_X(g^{-1}(y))\right]$$

Differentiating:

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

The slide also expresses this as:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

2.3 Case 2: $g(x)$ Monotonically Decreasing

Step S1: Compute CDF

$$F_Y(y) = \Pr(Y \leq y)$$

$$= \Pr(g(X) \leq y)$$

Because $g(x)$ is decreasing:

$$= \Pr(X \geq g^{-1}(y))$$

$$= 1 - F_X(g^{-1}(y))$$

Step S2: Differentiate

The slide shows the final expression in boxed form:

$$\boxed{f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|_{x=g^{-1}(y)}}}$$

The slide explicitly notes:

> S3: Change the limits for y

This step adjusts the support accordingly.

---

3. Example: Transformation of a Single Random Variable

3.1 Given

$$X \sim \text{Uniform}(-1, 1)$$

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

The PDF of X is given:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

3.2 Solution

From:

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Invert:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Differentiate:

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Using the transformation formula:

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}$$

The support shown:

$$-1 < y < 1$$

---

4. Function of Two Random Variables

4.1 Setup

$$Z = X + Y$$

The slide lists the following tasks:

1. Find PDF of Z, $f_Z(z)$
2. If X and Y are independent
3. Let $X \sim N(0,1)$ and $Y \sim N(0,1)$, prove $Z \sim N(0,2)$
4. If X and Y are exponential RVs with parameter λ, find $f_Z(z)$

---

5. Derivation of PDF of $Z = X + Y$

5.1 CDF Method

$$F_Z(z) = \Pr(Z \leq z)$$

$$= \Pr(X + Y \leq z)$$

The region corresponds to:

$$x \leq z - y$$

Thus:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

5.2 Leibniz Rule (as given on slide)

If:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

Then:

$$\frac{d}{dx} G(x) = \frac{d}{dx} b(x)\, h(x, b(x)) - \frac{d}{dx} a(x)\, h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)\, dy$$

The slide marks:

- Upper limit term
- Lower limit term
- Integral term

5.3 Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Applying the rule as shown:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y)\, dy$$

The slide also shows the equivalent "vertical strip" form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx}$$

---

6. Independent Case

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Therefore:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

The slide labels this:

> Convolution Integral

---

7. Normal Example

Given:

$$X \sim N(0,1), \quad Y \sim N(0,1)$$

PDFs shown:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

The algebra shown on slides leads to:

$$= \frac{1}{\sqrt{2\pi \cdot 2}} e^{-z^2/(2\cdot 2)}$$

Thus:

$$\boxed{Z \sim N(0,2)}$$

---

8. Exponential Example

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} u(x) u(z-x)\, dx$$

The slide specifies:

$$u(x)u(z-x) = \begin{cases} 1, & 0 < x < z \\ 0, & \text{otherwise} \end{cases}$$

Thus:

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \lambda^2 z e^{-\lambda z}$$

---

9. Example: $Z = X - Y$

$$F_Z(z) = \Pr(Z \leq z)$$

$$= \Pr(X - Y \leq z)$$

The region shown:

$$x \leq z + y$$

Thus:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y)\, dx\, dy$$

Then:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

The slide notes:

> Same procedure, $Z = X + Y$

---

10. Example: Ratio

$$Z = \frac{X}{Y}$$

$$F_Z(z) = \Pr\left(\frac{X}{Y} \leq z\right)$$

The slide distinguishes cases:

- $Y > 0$
- $Y < 0$

Regions are drawn accordingly.

11. Example: Radial Transformation

$$R = \sqrt{X^2 + Y^2}$$

The slide states:

$$f_R(r) = ?$$

(No further derivation shown.)

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LECTURE SCRIBE

Transformation of RVs + Function of Two RVs + Example

CSE400 — February 10, 2026

Dhaval Patel, PhD

---

1. Transformation of Random Variables

1.1 Topic Introduction

The lecture introduces:

1. **Transformation of Random Variables**
   Learning of transformation techniques for random variables.

2. **Function of Two Random Variables**

   Joint transformations and derived distributions.

3. **Illustrative Example**

   Detailed derivation for the case: $Z = X + Y$

1.2 Assumption

The slide explicitly states:

$$\textbf{Assumption: } f_X(x) \to \text{PDF of } X \text{ is known a priori}$$

The key objective highlighted:

$$\boxed{\text{How to find PDF of new transformed RV, } f_Y(y)?}$$

---

2. Function of One Random Variable

2.1 Setup

Let:

$$Y = g(X)$$

The slide distinguishes two cases:

- (+ve) Monotonically increasing
- (–ve) Monotonically decreasing

A graphical illustration is shown for both cases.

2.2 Case 1: $g(x)$ Monotonically Increasing

Step S1: Compute CDF

$$F_Y(y) = \Pr(Y \leq y)$$

$$= \Pr(g(X) \leq y)$$

Since $g(x)$ is increasing:

$$= \Pr(X \leq g^{-1}(y))$$

$$= F_X(g^{-1}(y))$$

Step S2: Differentiate with respect to y

$$f_Y(y) = \frac{d}{dy}\left[F_X(g^{-1}(y))\right]$$

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

The slide also writes:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

2.3 Case 2: $g(x)$ Monotonically Decreasing

Step S1: Compute CDF

$$F_Y(y) = \Pr(Y \leq y)$$

$$= \Pr(g(X) \leq y)$$

Since $g(x)$ is decreasing:

$$= \Pr(X \geq g^{-1}(y))$$

$$= 1 - F_X(g^{-1}(y))$$

Step S2: Differentiate

The slide presents the final expression:

$$\boxed{f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|_{x=g^{-1}(y)}}}$$

Step S3

Explicit note:

> Change the limits for y

---

3. Example: Transformation of a Single RV

3.1 Given

$$X \sim \text{Uniformly distributed RV} \sim (-1, 1)$$

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

PDF of X:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

3.2 Solution

From:

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Invert:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Differentiate:

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

Using the transformation formula:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}$$

Support shown on slide:

$$-1 < y < 1$$

---

4. Function of Two Random Variables

4.1 Setup

$$Z = X + Y$$

The slide lists:

(i) Find PDF of Z, $f_Z(z)$
(ii) $f_Z(z)$, if X & Y are independent
(iii) Let $X \sim N(0,1)$ & $Y \sim N(0,1)$, Prove $Z \sim N(0,2)$
(iv) If X & Y are exponential distribution RVs with parameter λ, find $f_Z(z)$

5. Derivation of $f_Z(z)$ for $Z = X + Y$

5.1 CDF

$$F_Z(z) = \Pr(Z \leq z)$$
$$= \Pr(X + Y \leq z)$$

Region (from diagram):

$$x \leq z - y$$

Thus:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

5.2 Leibniz Rule (as written)

If:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

Then:

$$\frac{d}{dx} G(x) = \frac{d}{dx} b(x)\, h(x, b(x)) - \frac{d}{dx} a(x)\, h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)\, dy$$

(Upper limit term, lower limit term, integral term are indicated on slide.)

5.3 PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Applying Leibniz rule as shown:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z-y, y)\, dy$$

Equivalent vertical strip form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z-x)\, dx}$$

---

6. Independent Case

If X & Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx}$$

The slide labels this:

> Convolution Integral

---

7. Normal Example

Given:

$$X \sim N(0,1), \quad Y \sim N(0,1)$$

PDFs shown:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

Algebra shown in slides leads to:

$$= \frac{1}{\sqrt{2\pi \cdot 2}} e^{-z^2/(2\cdot 2)}$$

Therefore:

$$\boxed{Z \sim N(0,2)}$$

---

8. Exponential Example

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$
$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} u(x) u(z-x)\, dx$$

The slide specifies:

$$u(x)u(z-x) = \begin{cases} 1, & 0 < x < z \\ 0, & \text{otherwise} \end{cases}$$

Thus:

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$
$$= \lambda^2 z e^{-\lambda z}$$

---

9. Example: $Z = X - Y$

$$F_Z(z) = \Pr(Z \leq z)$$
$$= \Pr(X - Y \leq z)$$

Region from slide:

$$x \leq z + y$$

Thus:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y)\, dx\, dy$$
$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Slide note:

> Same procedure, $Z = X + Y$

---

10. Example: Ratio

$$Z = \frac{X}{Y}$$
$$F_Z(z) = \Pr\left(\frac{X}{Y} \leq z\right)$$

The slide distinguishes:

- Case: $Y > 0$
- Case: $Y < 0$

Region illustrated accordingly.

---

11. Example: Radial Transformation

$$R = \sqrt{X^2 + Y^2}$$

The slide asks:

$$f_R(r) = ?$$

(No further derivation shown.)

End of Lecture Scribe

Generate a lecture scribe intended to serve as exam-oriented reference material.

Use ONLY the provided lecture slides as the authoritative context.
Do NOT use, infer, recall, or rely on any textbook, external

sources, prior training,
or general domain knowledge. If a definition, assumption, notation, theorem, proof
step, derivation, example, or numerical value does not explicitly appear in the lecture
slides, it must NOT be included.

The scribe must be sufficiently detailed so that a student could rely on it as the
primary revision document for a closed-notes or reading-based exam, without needing
to consult any other material.

While writing the scribe, reason step by step through the lecture content exactly as
presented in the slides. The reasoning must be explicit but strictly limited to what
is shown in the slides.

Follow the lecture sequence and organize the scribe into clearly labeled sections and
subsections that mirror the structure of the lecture. Ensure that every major topic,
subtopic, definition, result, and example introduced in the slides appears as its own
clearly identifiable section or subsection in the scribe.

Explicitly and systematically do the following:

1. For each concept introduced in the lecture, create a dedicated subsection that:
   - states the definition or statement exactly as given in the slides,
   - explains it step by step using only the wording, emphasis, and examples shown
     on the slides,
   - clearly distinguishes related concepts when the lecture explicitly does so.

2. When the lecture introduces properties, axioms, rules, corollaries, propositions,
   or results, present each one separately, with its exact mathematical form and
   accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if the steps are explicitly
   written on the slides. Do NOT fill in missing steps, add justifications, or extend
   arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
   full detail:
   - define all quantities and events exactly as done in the lecture,
   - show every intermediate step that appears on the slides,

- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major
concept must be written out completely, with the same level of detail as in the
lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow,
ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory,
or generalizations. If a proof, derivation, or explanation is omitted or only partially
shown in the slides, present it only to that same extent.

LECTURE SCRIBE

Transformation of RVs + Function of Two RVs + Example

CSE400

Dhaval Patel, PhD

February 10, 2026

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1. Transformation of Random Variables

1.1 Lecture Outline (as presented on slide)

1. **Transformation of Random Variables**

   Learning of transformation techniques for random variables.

2. **Function of Two Random Variables**

   Joint transformations and derived distributions.

3. **Illustrative Example**

   Detailed derivation for the case:

$$Z = X + Y$$

1.2 Key Objective

The slide explicitly states:

$$\boxed{\text{How to find PDF of new transformed RV, } f_Y(y)?}$$

1.3 Assumption

The slide specifies:

$$\textbf{Assumption: } f_X(x) \to \text{PDF of } X \text{ is known a priori}$$

Thus, throughout the transformation procedure, $f_X(x)$ is assumed known.

---

2. Function of One Random Variable

2.1 Definition

Let:

$$Y = g(X)$$

The slide explicitly separates two cases:

- (+ve) Monotonically increasing
- (–ve) Monotonically decreasing

Both cases are handled separately.

---

3. Case 1: $g(x)$ Monotonically Increasing

3.1 Step S1 – Compute CDF

$$F_Y(y) = \Pr(Y \leq y)$$

Since $Y = g(X)$:

$$= \Pr(g(X) \leq y)$$

Because $g(x)$ is increasing:

$$= \Pr(X \leq g^{-1}(y))$$

Therefore:

$$F_Y(y) = F_X(g^{-1}(y))$$

3.2 Step S2 – Differentiate w.r.t. y

$$f_Y(y) = \frac{d}{dy}\left[F_X(g^{-1}(y))\right]$$

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

The slide also writes the equivalent form:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

---

4. Case 2: $g(x)$ Monotonically Decreasing

4.1 Step S1 – Compute CDF

$$F_Y(y) = \Pr(Y \leq y)$$

$$= \Pr(g(X) \leq y)$$

Since $g(x)$ is decreasing:

$$= \Pr(X \geq g^{-1}(y))$$

$$= 1 - F_X(g^{-1}(y))$$

4.2 Step S2 – Differentiate

The slide presents the final expression:

$$\boxed{f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|_{x=g^{-1}(y)}}}$$

4.3 Step S3

The slide explicitly notes:

> Change the limits for y

---

5. Example: Single Variable Transformation

5.1 Given

$$X \sim \text{Uniformly distributed RV } \sim (-1, 1)$$

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

The PDF of X is given:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

5.2 Solution

From:

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Invert:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Differentiate:

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Using transformation formula:

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}$$

Support shown:

$$-1 < y < 1$$

---

6. Function of Two Random Variables

6.1 Setup

$$Z = X + Y$$

The slide lists:

(i) Find PDF of Z, $f_Z(z)$

(ii) $f_Z(z)$, if X & Y are independent

(iii) Let $X \sim N(0,1)$ & $Y \sim N(0,1)$, Prove $Z \sim N(0,2)$

(iv) If X & Y are exponential distribution RVs with parameter λ, find $f_Z(z)$

---

7. Derivation of PDF of $Z = X + Y$

7.1 Step 1 – CDF

$$F_Z(z) = \Pr(Z \leq z)$$
$$= \Pr(X + Y \leq z)$$

Region from slide:

$$x \leq z - y$$

Thus:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

---

8. Leibniz Rule (As Presented)

If:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

Then:

$$\frac{d}{dx} G(x) = \frac{d}{dx} b(x)\, h(x, b(x)) - \frac{d}{dx} a(x)\, h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)\, dy$$

The slide identifies:

- Upper limit term
- Lower limit term
- Integral term

9. PDF of Z

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Applying the rule as shown:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y)\, dy$$

Equivalent vertical strip form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx}$$

10. Independent Case

If X & Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

The slide labels:

> Convolution Integral

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Given:

$$X \sim N(0,1), \quad Y \sim N(0,1)$$

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Therefore:

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Region shown:

$$x \leq z + y$$

Thus:

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$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

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$$Z = \frac{X}{Y}$$

$$F_Z(z) = \Pr\left(\frac{X}{Y} \leq z\right)$$

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- $Y < 0$

Region drawn accordingly.

15. Example: Radial Transformation

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